

Online Predictive Motion Control of Redundant Robots via Model Predictive Control and Full Null Space Parameterization

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Abstract—This paper presents an online control method for redundant robots, designed to predictively prevent future violations of additional objectives while maintaining precise end-effector tracking. In this paper, we propose to use full null space parametrization (FNSP) to determine the optimal linear combination of null space basis vectors and use model predictive control (MPC) to optimize future control sequences for generating joint commands with foresight. Another critical aspect of this work is the online implementation. Analytical gradients of the formulated nonlinear programming (NLP) problem are derived to improve computational speed, and a retained rolling scheme is introduced within the MPC framework to reduce optimization dimensions. The proposed method is validated through tests on a 4-DOF planar manipulator and a 7-DOF Franka Panda robot, with numerical results demonstrating its efficiency.

I. INTRODUCTION

Leveraging the advancements in collaborative industry and control systems, redundant robots, which possess more degrees of freedom than the minimal required for the task space, offer enhanced capabilities, enabling the simultaneous handling of primary tracking tasks and safety-related additional tasks, such as collision avoidance, joint limits enforcement, and singularity avoidance. Therefore, effectively handling redundancy in real time is crucial. Nevertheless, global redundancy exploitation across the self-motion manifold is typically computationally intensive and lacks the responsiveness needed for safety-critical scenarios. Online control of redundant robots generally relies on the inversion of task kinematics, using local redundancy resolution to select the joint command within the null space of the task Jacobian.

One of the commonly used redundancy resolutions is the Gradient Projection Method (GPM) [1], where the end-effector velocity command is mapped into a minimal-norm joint command using the Moore–Penrose pseudoinverse of the task Jacobian [2]. The gradient of an objective function is projected into the null space of Jacobian, allowing for the optimization of the objective without compromising the primary task during motion execution. Similarly, many current state-of-the-art online approaches typically yield locally optimal solutions. They drive the robot configuration toward the descent objective with minimal motion effort, making them seemingly appropriate for each step. However, from a long-sighted horizon of continuous execution, reacting solely

through an instantaneous state can lead to global failures. For instance, the robot reaches the limit only to discover that the configuration satisfying the constraints is significantly misaligned with the current. Consequently, it is necessary to adopt predictable control strategies, enabling robots to balance immediate performance with longer-term rewards.

Considerable work has been invested in developing predictable control algorithms for robots [3]–[5]. [6] proposes a predictive inverse kinematics (IK) approach formulated as the quadratic programming (QP), which optimizes higher-order differential information by predicting future joint trajectories using Jacobian estimation. The problem of incorporating future information into the control scheme can be effectively addressed using Model Predictive Control (MPC) [7]. MPC solves a sequence of control variables over current and future time steps of a constrained optimization problem at each update. [8] proposes a predictive IK approach for redundant and underactuated robotic systems using MPC, utilizing a discrete linear double integrator for prediction and incorporating tracking error into the objective function, simplifying the problem into a QP. However, this simplification leads to a lack of strict enforcement of tasks. [9] uses a nonlinear model predictive control (NMPC) strategy to avoid emerging obstacles with task consistency. However, the computational cost associated with solving nonlinear MPC remains non-negligible. For real-time applications, [10] proposes to reduce Cartesian tasks to linear constraints and solve redundancy online using MPC. [11] further develop the NMPC with a projection-based redundancy resolution. By deriving analytical gradients for numerical optimization and implementing a multi-threaded execution scheme, real-time performance is enhanced.

This paper presents the Predictive Motion Control (PMC) for redundant robots, which facilitates locally optimal control strategies to generate trajectories for long-lasting guaranteed additional objectives predictively. The first contribution of this paper is to propose a more robust framework for predictive control. We use the Full Null Space Parameterization (FNSP) within an MPC framework to solve an optimization problem for the foresighted selection of null space motion. Though the GPM is relatively straightforward, it cannot guarantee motion constraints, and numerical problems may arise from the *global incompatibility* related to projection as pointed out in [12]. In contrast, FNSP exploits a full set of base vectors spanning the null space to provide a versatile and intuitive formulation [13], which will be shown in this paper. Additionally, the real-time performance of the method is considered. Analytical gradients of the

This work was supported by National Natural Science Foundation of China under Grant 12272266.

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nonlinear programming problem are derived using the direct shooting method, which significantly improves the speed of optimization solving. Furthermore, a retained rolling scheme is proposed to reduce optimization dimensions within MPC. The presented NMPC framework applies to general nonlinear objectives and can be implemented with a high control frequency for real-time computation. The effectiveness of the proposed method is thoroughly validated through simulations, including a limit avoidance test with a 4-DOF planar manipulator and a collision avoidance test with a 7-DOF Franka Emika Panda robot.

II. REDUNDANCY RESOLUTION

Consider an n -DOF robot with joint coordinates $\mathbf{q} \in \mathbb{R}^n$, $\mathbf{x} \in \mathbb{R}^m$ the vector of variables describing a m -dimensional task. Joint velocity command using the GPM by Liegeois [1] can be generated as:

$$\dot{\mathbf{q}} = \mathbf{J}^\dagger \dot{\mathbf{x}} - \alpha(\mathbf{I} - \mathbf{J}^\dagger \mathbf{J}) \left(\frac{\partial h(\mathbf{q})}{\partial \mathbf{q}} \right) \quad (1)$$

where $\mathbf{J}^\dagger$ is the Moore–Penrose pseudoinverse of the task Jacobian $\mathbf{J}(\mathbf{q}) \in \mathbb{R}^{m \times n}$, it calculates a minimal-norm mapped joint command w.r.t. desired velocity $\dot{\mathbf{x}}$ of the end-effector. GPM utilizes redundancy to prefer additional task by projecting the gradient of the objective function $h(\mathbf{q})$ into the null space of Jacobian through a $m \times m$ projection matrix $\mathbf{N} = \mathbf{I} - \mathbf{J}^\dagger \mathbf{J}$. α is a positive gain factor.

Although GPM is a relatively straightforward method, it is difficult to provide a rigorous and versatile framework for optimization. Since the gradients used as control variables are projected via a projection matrix, the selection of base vectors in the null space is omitted. This results in the projected one-dimensional subspace lacking a clear representation, and thus the optimal combination of null space vectors is not guaranteed. Another drawback is that the numerical stability problem may arise as described in [14], since the projected vector does not actually represent the gradient of the potentials.

We present the better redundant resolution alternative FNSP in the proposed method, as introduced in [13]. The self-motion is formulated as the linear combination of the null-space base vectors:

$$\dot{\mathbf{q}} = \mathbf{J}^\dagger \dot{\mathbf{x}} + \mathbf{G}\mathbf{u} \quad (2)$$

where $\mathbf{u} \in \mathbb{R}^{n-m}$ is the control variables, $\mathbf{g}_i \in \ker(\mathbf{J})$, $i = 1, \dots, n - m$ are the null space base vectors. $\mathbf{G}$ can be calculated from the SVD performed on $\mathbf{J}$:

$$\mathbf{J} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T \quad (3)$$

$$\mathbf{G} = \mathbf{V}_{null} \quad (4)$$

where $\mathbf{V}_{null}$ consists of the columns of $\mathbf{V}$ associated with zero singular values in the SVD.

Compared to GPM, which has been shown in [13], the gradient projection is an evenly weighted preference for all null space components w.r.t. an objectives-related quadratic criterion. In contrast, the search region of the FNSP is generalized to span the entire local solution space, which

provides a feasible framework for predictive motion control [15].

III. PREDICTIVE MOTION CONTROL

A. MPC Model with FNSP

The proposed PMC, based on MPC, solves an optimal control problem that includes the states of future sequences within a predictive horizon time T_{ph} from the current time t_0 , and computes the control variables at the current update step. By setting the control variables directly as the null space command $\mathbf{u}$ in (2), the implementation of MPC is formulated as the following discrete nonlinear program (DNP).

$$\begin{aligned} \mathbf{u}_{1:n_p} &= \arg \min_{\mathbf{u}} \sum_{k=1}^{n_p-1} L_k(\mathbf{q}_k, \tilde{\mathbf{u}}_k) \\ \text{s.t. } \forall k &\in [1, \dots, n_p] \\ \dot{\mathbf{q}}_k &= \mathbf{f}(\mathbf{q}_k, \dot{\mathbf{x}}_k, \mathbf{u}_k) \\ \mathbf{q}_{k+1} &= \mathbf{q}_k + \dot{\mathbf{q}}_k t_{ps} \\ \mathbf{u}_k &\in \mathbf{U}_{tol}; \mathbf{q}_k \in \mathbf{Q}_{safe}; \dot{\mathbf{q}}_k \in \dot{\mathbf{Q}}_{safe}; \ddot{\mathbf{q}}_k \in \ddot{\mathbf{Q}}_{safe} \end{aligned} \quad (5)$$

where horizon of MPC is discretized into n_p prediction steps of time size t_{ps} . $\tilde{\mathbf{u}}_k = [\mathbf{u}_1, \dots, \mathbf{u}_{k-1}]^T$ is the control variables set containing from k -th step and earlier. $\mathbf{f}(\mathbf{q}_k, \dot{\mathbf{x}}_k, \mathbf{u}_k)$ derives $\mathbf{q}_{k+1}$ following (2). $\mathbf{U}_{tol}$ is the tolerance range of allowed inputs. PMC calculates the control variables $\mathbf{u}_{1:n_p}$ distributed within T_{ph} , and use $\mathbf{u}_1 \in \mathbb{R}^{n-m}$ as the control input of current update.

The formulated NLP includes nonlinear constraints and a cost function $L_k(\mathbf{q}_k, \tilde{\mathbf{u}}_k)$ that supports both general linear and nonlinear objective functions $h(\mathbf{q})$. Note that the objectives should allow for the calculation of their gradients to facilitate online execution. The detailed implementation is presented in Sec. IV.

B. Cost Function with Constraint Penalty

The essential goal of predictable kinematic control is to foresee and avoid any violation of the additional task. However, almost all safety-related objectives, mainly such as obstacle avoidance, are transformed into optimizing the distance to the tolerant boundary. Lack of hard constraints to handle the potential objective failure, especially in MPC, we want the optimization sequence to have the “attention”-that is, the emerging boundary breach predicted in the future can affect the selection of prior configuration and initiate a necessary transition.

To this end, we introduce a simple penalty term into the cost function

$$\begin{aligned} \tilde{L}_k &= h_k(\mathbf{q}_k, \tilde{\mathbf{u}}_k) \\ &+ \gamma \exp(\max(\mathbf{0}, \max(\mathbf{l}_k - \mathbf{l}_{\max}), \min(\mathbf{l}_{\min} - \mathbf{l}_k))) \end{aligned} \quad (6)$$

where $\mathbf{l}_k(\mathbf{q}_k, \tilde{\mathbf{u}}_k)$ is the possible vector representation w.r.t. the hard limits of scalar objective h_k , if the boundary cannot be simplified to a scalar form. γ is a large positive gain to reinforce the penalty.

C. Shaped Velocity Constraints

The control variables of the PMC are velocity-based, thus the joint angle and acceleration constraints are shaped into the joint velocity constraints using finite difference methods. The acceleration is approximated with a backward difference scheme, and the joint position and velocity formulations for $t \in [t_k, t_{k+1}]$ are:

$$\dot{q}(t) = \dot{q}_k + \frac{\dot{q}_k - \dot{q}_{k-1}}{\Delta t}(t - t_k) \quad (7)$$

$$q(t) = q_k + \dot{q}_k(t - t_k) + \frac{\dot{q}_k - \dot{q}_{k-1}}{2\Delta t}(t - t_k)^2 \quad (8)$$

where Δt is the sampling time duration. Then the joint range limits, velocity, and acceleration bounds are derived similarly to [16], the constrained on q_k can be shaped as follows (for the sake of brevity, only the maximum bound is shown):

$$\dot{Q}_{\max} = \min \left\{ \frac{2(Q_{\max} - q_k)}{3\Delta t} + \frac{\dot{q}_{k-1}}{3}, V_{\max}, A_{\max}\Delta t + \dot{q}_{k-1}, \sqrt{2A_{\max}(Q_{\max} - q_k)} \right\} \quad (9)$$

where the last term is the additional viability constraint that guarantees the joint position bound without exceeding the maximum acceleration [17].

IV. ONLINE IMPLEMENTATION

A. Analytical Gradients for Optimization Solver

The nonlinear model predictive control (NMPC) in PMC uses an interior-point nonlinear optimization solver. The analytical gradients of cost function and constraints are needed for fast implementation. The state-of-the-art numerical optimal control approaches have been benchmarked in [11]. In this subsection, we refer to the derivation for the preferred direct shooting method in [11] and present the analytical gradients for FNSP.

With the definition of the complete sequence of predictive horizon $\hat{q}, \hat{u}$ and the k -th step control history $\tilde{u}_k$

$$\hat{q} = [q_1, \dots, q_{n_p}]^T, \quad \hat{u} = [u_1, \dots, u_{n_p}]^T \quad (10a)$$

$$\tilde{u}_k = [u_1, \dots, u_{k-1}]^T. \quad (10b)$$

The analytical gradient of the cost function satisfies the following derivation:

$$\frac{\partial \tilde{L}(\hat{q})}{\partial \hat{u}} = \sum_{k=1}^{n_p-1} \left(\frac{\partial \tilde{L}_k(q_k, \tilde{u}_k)}{\partial u_k} \frac{\partial u_k}{\partial \hat{u}} + \frac{\partial \tilde{L}_k(q_k, \tilde{u}_k)}{\partial q_k} \frac{\partial q_k}{\partial \hat{u}} \right) \quad (11a)$$

$$\frac{\partial \tilde{L}_k(q_k, \tilde{u}_k)}{\partial u_k} \frac{\partial u_k}{\partial \hat{u}} = 0 \quad (11b)$$

where $h_k(q_k, \tilde{u}_k)$ and $l_k(q_k, \tilde{u}_k)$ in $\tilde{L}$ are both admit analytical derivatives w.r.t. q_k . To calculate the latter term in (11a), the sensitivity differential equations (SDEs) are used according to [18]. Let $q_{S,k}$ be the sensitivity w.r.t. to q_k , the

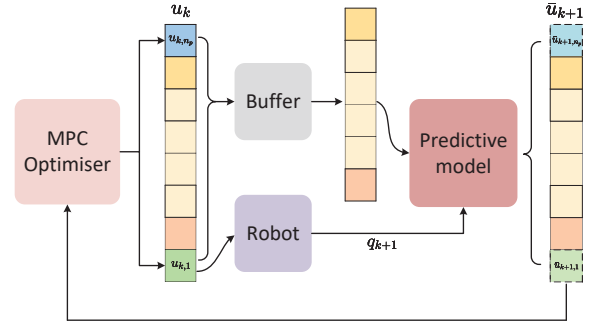


Fig. 1. Retained rolling implementation in MPC.

following recurrence relation is obtained:

$$q_{S,k+1} = \frac{\partial q_{k+1}}{\partial \hat{u}} = t_{ps} \frac{\partial \dot{q}_k}{\partial \hat{u}} + q_{S,k} \quad (12a)$$

$$\begin{aligned} \frac{\partial \dot{q}_k}{\partial \hat{u}} &= \frac{\partial \dot{q}_k}{\partial u_k} \frac{\partial u_k}{\partial \hat{u}} \\ &= \frac{\partial f(q_k, u_k)}{\partial u_k} \frac{\partial u_k}{\partial \hat{u}} + \frac{\partial f(q_k, u_k)}{\partial q_k} q_{S,k} \end{aligned} \quad (12b)$$

where $q_{S,1} = \frac{\partial q_1}{\partial \hat{u}}$ is 0 , $\frac{\partial u_k}{\partial \hat{u}} = [0, \dots, I_{n-m}, \dots, 0]^T$. Next, the partial derivatives w.r.t. $f(q_k, u_k)$ holds:

$$\frac{\partial f(q_k, u_k)}{\partial u_k} = G(q_k) \quad (13a)$$

$$\frac{\partial f(q_k, u_k)}{\partial q_k} = \frac{\partial J^\dagger(q_k)}{\partial q_k} \dot{x} + \frac{\partial G(q_k)}{\partial q_k} u_k \quad (13b)$$

$$\begin{aligned} \frac{\partial J^\dagger(q_k)}{\partial q_{k,j}} &= \left(\frac{\partial J_k}{\partial q_{k,j}} \right)^T (J_k J_k^T)^{-1} - J_k^T [\\ & (J_k J_k^T)^{-1} \left(\frac{\partial J_k}{\partial q_{k,j}} J_k^T + J_k \left(\frac{\partial J_k}{\partial q_{k,j}} \right)^T \right) (J_k J_k^T)^{-1}] \end{aligned} \quad (13c)$$

$$\frac{\partial V(q_k)}{\partial q_{k,j}} = (I - V V^T) \left(\frac{\partial J_k}{\partial q_{k,j}} \right)^T U \Sigma^+ \quad (13d)$$

$$\frac{\partial G(q_k)}{\partial q_{k,j}} = \left(\frac{\partial V(q_k)}{\partial q_{k,j}} \right)_{null} \quad (13e)$$

where the subscript j denotes the joint number, and $\frac{\partial J(q_k)}{\partial q_{k,j}}$ can be calculated on the geometric analysis of the kinematic chain of robots. In addition, the analytical gradients of nonlinear constraints are straightforward using the derivatives derived above.

B. Retained Rolling MPC

In online implementations, a high computation burden of NMPC can limit the control frequency. Although using analytical gradients has improved the optimization speed, the increased optimization dimensions required to support a sufficiently long predictive horizon cannot be overlooked [19]. To address this, we introduce sequential retention and receding horizon implementation within the MPC framework to reduce the optimization dimensions effectively.

In standard MPC, a sequence of control variables over a future time horizon is computed. However, only the first term $\mathbf{u}_{n,1}$ from $\hat{\mathbf{u}}_n = [\mathbf{u}_{n,1}, \dots, \mathbf{u}_{n,n_p}]$ is applied to the robot at the current update t_n . The overlapping sequences in the receding horizon are recalculated at each time step. Here, n in $\mathbf{u}_{n,k}$ denotes the execution step and k denotes the prediction step. We approximate the current predictive control sequence $\hat{\mathbf{u}}_n$ as a prediction of the future state at the next control update. During updating, the calculated control sequences $\hat{\mathbf{u}}_n$ are stored in a buffer, with $\mathbf{u}_{n,1}$ being inputted to the robotic system and output the new state $\mathbf{q}_{n+1}$. At each rolling horizon in the MPC, only need to update the first control variable $\mathbf{u}_{n+1,1}$ and the last term $\mathbf{u}_{n+1,n_p}$ within the prediction interval, while the intermediate control variables are retrieved directly from the buffer. The retained sequences will be aligned over the time series to ensure consistency between the predictive model and the system dynamics. The rolling scheme can be formulated as follows:

$$\bar{\mathbf{u}}_{n+1} = [\mathbf{u}_{n+1,1}, \mathbf{u}_{n,2}, \dots, \mathbf{u}_{n,n_p-1}, \mathbf{u}_{n+1,n_p}] \quad (14)$$

$$\mathbf{u}_{n+1,\{1,n_p\}} = \arg \min_{\mathbf{u}} \sum_{k=1}^{n_p-1} L_{n+1,k}(\mathbf{q}_{n+1,k}, \bar{\mathbf{u}}_{n+1}) \quad (15)$$

where $\bar{\mathbf{u}}_{n+1}$ represents the control sequences at the update time t_{n+1} . The rolling horizon control structure is shown in Fig. 1. This implementation allows for an extended prediction horizon, with the initial control steps serving as a warm-up that is precomputed offline. However, the trade-off between the length of the prediction horizon and the accuracy of predicted states should be considered, as biases caused by the approximation of the predictive model may arise when the horizon extends too far.

V. RESULTS

To evaluate the performance of our method, we conducted two tests: one on a 4-DOF planar manipulator and another on a 7-DOF manipulator, each with different additional objectives. The proposed method was compared with commonly used local control approaches, demonstrating its computational efficiency and feasibility.

A. Planar Manipulator Joint Limit Avoidance

In this test, we ran a simulation of a 4-DOF planar manipulator. Each segment of the manipulator has a length of 0.1 m, with the base fixed at the origin and all joints rotational. The manipulator is executed to move along a straight line following joint range limits

$$\mathbf{Q}_{\max} = -\mathbf{Q}_{\min} = \left[\frac{\pi}{2} \quad \frac{\pi}{2} \quad \frac{\pi}{2} \quad \frac{\pi}{2} \right]^T \text{ (rad)}$$

The manipulator moves from the initial configuration $\mathbf{q}_0 = [-0.85, 1.25, 0.85, 0.73]^T \text{ (rad)}$ w.r.t. the initial position of end-effector $\mathbf{x}_0 = [0.15, 0.15]^T \text{ (m)}$, and stops at $\mathbf{x}_{\text{end}} = [-0.07, 0.07]^T \text{ (m)}$. The objective function is defined as the distance from the center of the joint range limits, given by $h(\mathbf{q}_k) = \frac{1}{2} \mathbf{q}^T \mathbf{q}$ and $l(\mathbf{q}_k) = \mathbf{q}$. The upper and lower punishment boundaries defined as $l_{\max} = \mathbf{Q}_{\max}$, $l_{\min} = \mathbf{Q}_{\min}$.

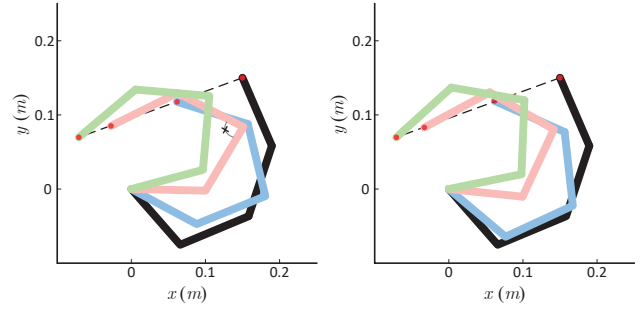


Fig. 2. Snapshots of the 4-DOF planar manipulator moving along a straight line. The pink presents the peak of joint motion. The left image using GPM results in a limit violation at the third joint; the Right image uses PMC.

TABLE I
COMPARISON OF IMPLEMENTATION TIME

	Nodes of path	Implementation time (s)
NMPC	236	588.39 ± 10.01
NMPC with AG	236	20.43 ± 1.05
PMC	236	8.95 ± 0.58

The NLP problem of the proposed scheme is solved using the interior-point solver of MATLAB R2020a on a 64-bit AMD Ryzen7 6800H (3.2GHz) CPU with 16GB of RAM. The predictive horizon is set $n_p = 20$ with each time step $t_{ps} = 0.1 \text{ s}$.

To demonstrate the performance of the proposed method, we evaluate the comparisons against commonly used local control approaches, specifically the Gradient Projection Method (GPM) and Pseudo-Inverse Control (PIC), in which the latter does not account for any additional objectives. The same objective function is applied in GPM for a fair comparison.

The simulation results for the end-effector trajectories and joint commands are shown in Fig. 2-5. It can be observed that each step in PIC is executed with minimal joint velocity. Notably, the second joint, closest to the boundary, consistently tends to move towards violating the limits. This aligns with our expectations, as PIC does not make any efforts to handle joint limits. The comparison of GPM with the proposed method is noteworthy. Despite the fact that the same objective has been added to the GPM, the joint range limits are still not guaranteed. It can be concluded that while GPM optimizes the motion of the second joint, its lack of predictive capability becomes evident when the joints approach the boundary of limits, making an effective transition difficult. In contrast, the proposed method is aware of the future risks and takes an effective transition in advance.

Table I shows the comparison of average implementation times for NMPC using different stages of techniques with ten runs. Where NMPC stands for instead of using the analytical gradients, the interior-point solver uses an automatic finite difference method. PMC is the published version with analytical gradients (AG) and the retained rolling mechanism. The result shows that PMC reduces computational time by

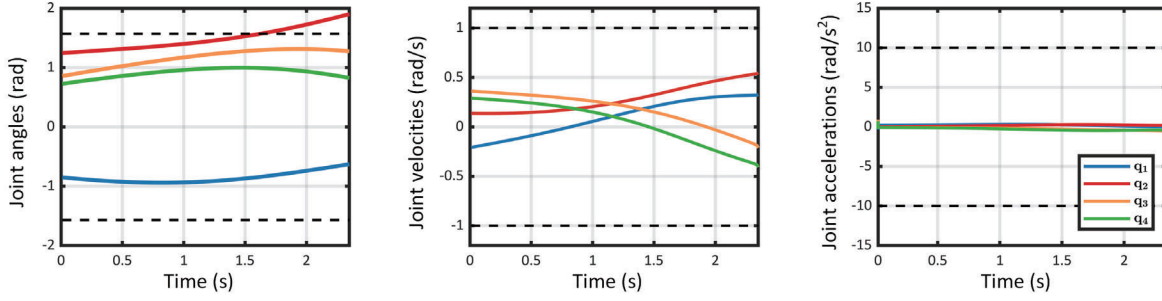


Fig. 3. Joint angle, velocity, acceleration results of the planar manipulator using PIC.

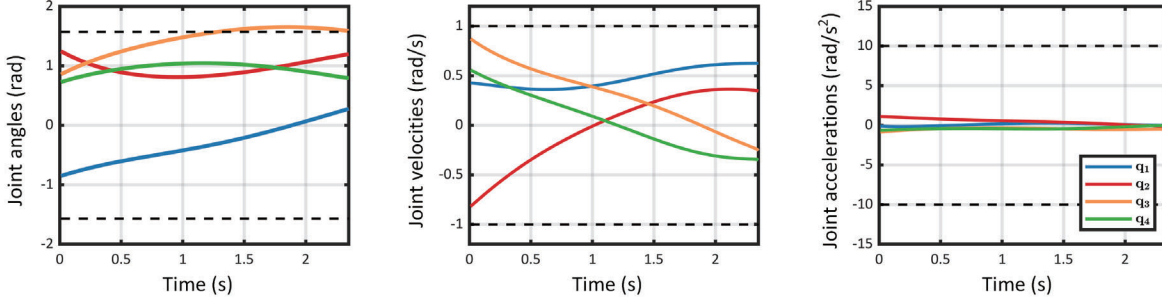


Fig. 4. Joint angle, velocity, acceleration results of the planar manipulator using GPM.

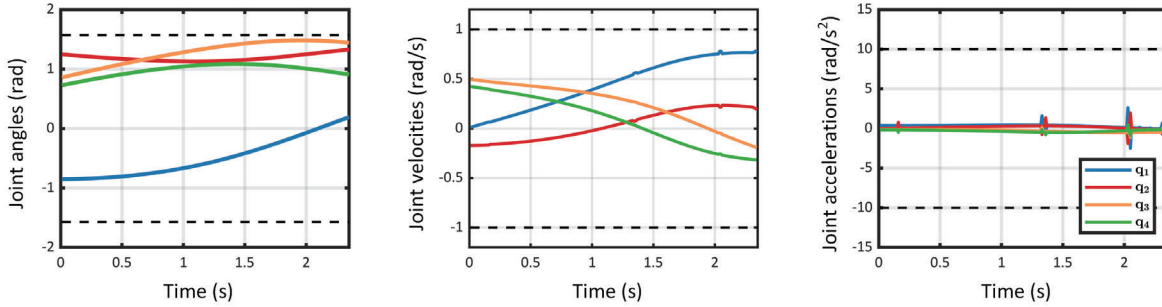


Fig. 5. Joint angle, velocity, acceleration results of the planar manipulator using PMC.

98.48% compared to the original NMPC, demonstrating the efficiency of fast implementation.

B. 7-DOF Manipulator Collision Avoidance

In this test, we deploy the proposed algorithm on a Franka Emika Panda manipulator. The optimization problem in the proposed PMC is solved using the *NLopt* nonlinear optimization library [20] in C++. The online control is implemented through the Robot Operating System (ROS) interface provided by the *Franka Control Interface* (FCI), and Gazebo is used as the virtual robotic validation environment. Note that the required control sampling time in *libfranka* is 1 ms. When the algorithm cannot be executed at the required frequency, a null joint command is passed to allow the necessary wait time.

The task in this section is end-effector of the manipulator is executed to track a spiral line which is given as:

$$\begin{cases} \dot{x}_e = 0.4 \cos(\pi t - \pi/2) \\ \dot{y}_e = 0.8 \sin(\pi t - \pi/2) \\ \dot{z}_e = 0.02 \end{cases} \quad (16)$$

The manipulator tracks position but not orientation, thus the redundancy degree is 4. As shown in Fig. 6, it operates under a table, with the fourth joint positioned dangerously close to the surface. Selecting appropriate null space commands is critical to avoid a potential “head knocks.” The objective function for null space tasks, used across all comparison methods, is defined as the negative of the collision distance. This distance during execution is calculated using the Cartesian coordinates of the fourth joint, derived through forward kinematics. The manipulator begins from the initial configuration $\mathbf{q}_0 = [0, 1.09, 0, -2.53, 0, 0.79, 0]^T$ (rad), corresponding to the end-effector’s initial position $\mathbf{x}_0 = [0.13, 0, 0]^T$ (m). For the PMC implementation, the predictive horizon is set to $n_p = 20$, with a step size of $t_{ps} = 0.1$ s.

Fig. 7 shows a comparison of the collision distances using the three methods. During the interval $[1, 1.5]$ (s), both GPM and PIC violate the safety collision limit. In contrast, PMC is able to react before 1 s, and effectively adjusts the collision distance in the opposite direction.

Additionally, we evaluated the control frequency achievable with the PMC. For a system with four degrees of redun-

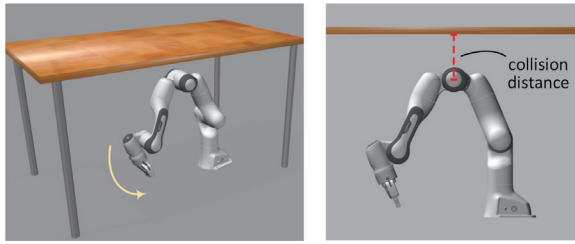


Fig. 6. Illustration of collision avoidance during tracking under a table, with a focus on the fourth joint to measure the most critical distance.

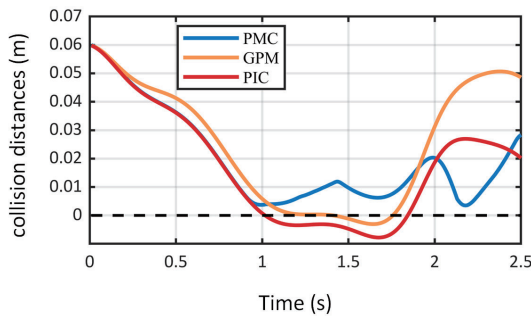


Fig. 7. Comparison of collision distances of joint trajectories generated by the three methods.

dancy, PMC successfully generates joint commands in real-time under general nonlinear objectives at a frequency of up to 100 Hz. When the redundancy is reduced to two degrees, the control frequency can be increased to a maximum of 300 Hz. In conclusion, using analytical gradients and the retained rolling implementation in the NMPC framework, as presented in this paper, enhances computational efficiency. These optimizations enable the potential application of NMPC in real-time scenarios, though the achievable performance will depend on the system's complexity and the available computational resources.

VI. CONCLUSION

This paper presents an online control method for redundant robots that, while inherently local, enables joint motion generation with foresight, effectively avoiding potential violations of additional tasks without interfering the end-effector tracking. This is achieved by predictive motion control, i.e., using MPC to simultaneously optimize current and future motion in the null space. This paper used FNSP to handle redundancy, providing a versatile framework for effective optimization. Additionally, the online capabilities of the proposed method are enhanced by using the derived analytical gradients of objective function and constraints within the formulated NLP. A retained rolling scheme within the MPC framework is also introduced to reduce optimization dimensions. Case studies have demonstrated that the proposed method can preemptively avoid emerging violations of additional tasks, while maintaining a considerable control frequency for general nonlinear objectives.

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